Causal Inference When the Map Changes: Partition Designs for Administrative Boundary Reforms

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Abstract

Governments routinely redraw internal maps by merging, splitting or reclassifying jurisdictions, handing every unit a bundle of treatments at once. This paper develops partition designs, which treat the published rule behind a new map as the assignment mechanism and sample the maps it allows with a reversible merge-split chain. Design-orthogonalised dose instruments are exactly unbiased, as implemented, for a convex average of one component's effects, and remove the cross-component contamination that recentring leaves. Besag–Clifford draws give tests that are exact for any chain length; reweighting them gives exact tests under any alternative design, and hence an exact sensitivity analysis for the design assumption. On Denmark's 2007 municipal reform, four standard estimators report a significant cost of larger jurisdictions yet fail pre-reform placebos; the design-based tests pass them and show the reform could not have detected an effect ten times larger. A placebo on Vietnam's 2025 consolidation confirms the pattern.

Keywords: administrative boundaries, municipal mergers, design-based inference, exposure mappings, contamination bias, randomization tests, sensitivity analysis, Markov chain Monte Carlo.

JEL codes: C12, C14, C21, C31, H77, R50.

1. Introduction

Governments redraw their internal maps far more often than economists study the act of redrawing. Hundreds of municipal amalgamations in Europe and Japan since 2000, documented among others by Blesse and Baskaran (2016) and Weese (2015), have each assigned whole populations to new governments on a single date. Yet the econometrics applied to these reforms is borrowed almost wholesale from designs built for a

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policy that is switched on in some places and left off in others.

Vietnam's reform of 2025 is a stark case. On 12 June 2025 the National Assembly adopted Resolution 202/2025/QH15, which cut the country's 63 provinces to 34 from 1 July, while a parallel reform abolished the district tier and folded 10,035 communes into 3,321 (National Assembly of Vietnam 2025; Government of Vietnam 2025). Twenty-nine provincial capitals lost their status overnight, and every commune in a merged province acquired a new capital, new neighbours inside its jurisdiction and, in most cases, a new distance to the seat of government.

Applied work on such reforms usually regresses an outcome on a "merged" or "lost capital" indicator interacted with a post-period dummy, clusters by old province and reports t-statistics. Three problems sit inside that routine. The rest of the paper takes them in turn.

The first problem is that a map change is not a binary treatment. It moves at least four things together for each unit: whether its capital survives, how far it is from its capital, how large its jurisdiction becomes and whether a border it sat on turns internal. A regression on one indicator loads all of them onto one coefficient, and a regression on all of them suffers the contamination bias characterised by Goldsmith-Pinkham, Hull and Kolesár (2024) whenever the components are jointly assigned.

The second problem is inference. Vietnam's reform merged 52 old provinces into 23 new ones, so clustering at the old-province level leaves roughly two dozen effective treated groups whose doses are tied to geography, a regime in which cluster-robust and wild-bootstrap methods are known to struggle (Cameron, Gelbach and Miller 2008; MacKinnon and Webb 2017). None of those methods uses the one piece of information a map reform hands the analyst for free: the published rule that produced the map.

The third problem is data. Statistical offices publish on the map in force, so post-reform outcomes appear only for the 34 new provinces while the treatment varies across the 63 old ones. National co-treatments compound the difficulty: the abolition of districts and the consolidation of communes touched every province, so the 11 "unchanged" provinces are not untreated controls. This paper handles the co-treatments and works with outcomes on a geography the reform does not redraw; recovering the effects of absorbed areas from statistics that exist only on the new map is the subject of a companion paper.

This paper proposes a single framework for these settings, which we call a partition design. A partition-change design, in our definition, is any reform in which a published rule maps an old partition of space into a new one, and a partition design treats that rule as the source of identifying variation in the spirit of design-based inference (Abadie, Athey, Imbens and Wooldridge 2020). The framework has four parts, summarised as a seven-step protocol in Figure 1.

The first part is identification. We code the criteria as a set of admissible maps, model the remaining discretion

with a low-dimensional assignment distribution, and construct, for every unit and every treatment component, an instrument equal to the realized dose minus its design projection on the unit's other doses. Theorem 1 shows that the resulting estimator is design-unbiased for a convex average of that component's effects, with no weight on the others, extending the recentring logic of Borusyak and Hull (2023) from a single exposure to a vector of jointly assigned exposures.

The second part is exact inference. Nothing in the resulting test requires the Markov chain to mix, because the parallel construction of Besag and Clifford (1989) makes the realized map exchangeable with the simulated ones under the design. Theorem 2 adds that a Berger–Boos correction keeps the test valid when the design parameters are estimated from the same map, and Proposition 3 shows that co-treatments the map does not move drop out of the estimating equation altogether, so no untreated control group is needed.

The third part is robustness. Proposition 4 shows that reweighting the simulated maps toward any other design gives exact, not approximate, tests; we use it for an exact sensitivity analysis of the design assumption, whose breakdown values we benchmark against how strongly the realized map selected on observed pre-reform trends, in the spirit of Rosenbaum (2002) and Cinelli and Hazlett (2020). Theorem 2(c') adds a studentised test that stays valid when the other components' effects are heterogeneous. The fourth part is evidence: a full application to Denmark's 2007 municipal reform with parish-level outcomes, a real-data placebo and Monte Carlo studies on Vietnam's geography, and an application that codes Vietnam's 2025 criteria, estimates the capital-choice rule and diagnoses the design.

The results are concrete, and they show why the design is needed. In Denmark, where 271 municipalities became 98 under a 20,000-inhabitant guideline, we rebuild the reform from official parish data for 1,637 parishes. Multi-dose regression, difference-in-differences with untouched municipalities as controls, triple differences and instrumental variables with the design-expected dose all conclude that joining a larger municipality slowed annual population growth, by 0.12 to 0.26 percentage points per log point of size (p between below 0.001 and 0.013), yet each finds significant "effects" of the reform's doses on growth before 2007. The design-based tests pass all six pre-reform placebos at the fitted design and reject nothing. They also report their own power: the design could not detect a size effect ten times the regression estimate, so the regressions' conclusion rests entirely on a parallel-trends assumption that the placebos reject. An exact sensitivity analysis flags internal borders as the component on which councils selected, and the protocol declines to interpret it. In Vietnam, the realized 2025 map lies between the 59th and 90th percentiles of 2,000 admissible maps on seven features, and the fitted capital rule predicts 78 percent of the actual capital choices. Regressing five characteristics fixed before the reform on its doses produces 5 significant "effects" out of 15 with clustered standard errors and 4 out of 15 with Conley errors; partition-design tests produce none, at every value of the compactness tilt and under every variant of the admissible set. In simulations with planted effects, multi-

dose regressions reject a true null in 45 to 47 percent of samples once untreated trends follow the design, and recentred instruments estimate a size effect of 0.40 at 0.17; the design-orthogonalised estimator is on target in every scenario, including one on Danish geography in which the true map comes from an independent long chain and the analyst uses a 20-step one; there recentring estimates a truly zero size effect at 0.32.

The paper connects three literatures that have so far developed separately. Graph-partition samplers from redistricting research (Fifield, Higgins, Imai and Tarr 2020; McCartan and Imai 2023) supply the computational engine, but there the map aggregates votes into seats, whereas here it defines the governments whose actions are the treatment. Design-based inference for shocks with non-random exposure (Borusyak and Hull 2023) and the contamination analysis of Goldsmith-Pinkham, Hull and Kolesár (2024) supply the identification logic, which we extend from vectors of shocks to partitions and from regression controls to unit-level design projections.

We also build on design-based difference-in-differences (Athey and Imbens 2022), on inference with few treated groups and approximate symmetry (Conley and Taber 2011; Ferman and Pinto 2019; Canay, Romano and Shaikh 2017), on Monte Carlo tests with nuisance parameters (Dufour 2006) and Markov chain significance tests (Besag and Clifford 1989; Chikina, Frieze and Pegden 2017; Howes 2023). Relative to exposure mappings and interference (Aronow and Samii 2017; Athey, Eckles and Imbens 2018), our exposures are vectors assigned jointly by one partition; relative to generalized propensity scores (Imbens 2000; Hirano and Imbens 2004), the design distribution of the whole dose vector is known from the rule rather than estimated; and relative to the design-based remedies of Goldsmith-Pinkham, Hull and Kolesár (2024), which reweight by known propensities of discrete treatments, design-orthogonalised instruments partial continuous, jointly assigned doses unit by unit. The empirical literature on amalgamations (Lassen and Serritzlew 2011; Hirota and Yunoue 2017; Egger, Koethenbuerger and Loumeau 2022) and on district splits (Bazzi and Gudgeon 2021) supplies the settings where these tools apply. Nothing in the theory is specific to maps. Theorems 1' and 2 and Proposition 4 apply whenever a known rule assigns units to groups and each unit's treatments are functions of its group, which covers electoral redistricting (Chikina, Frieze and Pegden 2017), the redrawing of school attendance zones and the definition of clusters in network experiments.

Section 2 describes the setting and the protocol. Section 3 sets up the framework. Section 4 shows what standard estimators average. Section 5 presents the design-orthogonalised estimator and Section 6 its inference and sensitivity analysis. Section 7 applies the full protocol to Denmark's 2007 municipal reform with parish micro-data, and Section 8 summarises the evidence from Vietnam's 2025 consolidation. Section 9 concludes and Appendix A gives the proofs. The Supplemental Appendix reports the Vietnamese placebo, simulations and design diagnostics in full (S1 to S4), shows how to apply the method to other reforms (S5), and documents the software and data (S6 and S7).

2. Setting and protocol

Vietnam's 2025 reform redrew the province tier and removed the district tier in one step. Eleven provinces were left untouched, and the remaining 52 were combined into 17 pairs and 6 triples, each keeping exactly one predecessor's capital (Figure 2). The commune tier was consolidated at the same time, so the post-reform state has two tiers instead of three.

Table 1. The reform at a glance. Sources: Resolution 202/2025/QH15; Government Resolutions 125/NQ-CP and 126/NQ-CP of 9 May 2025. The district count is approximate.

Element	Before 1 July 2025	After 1 July 2025	What it changes for a commune
Provinces	63	34 (11 unchanged, 23 formed by merger)	which government it answers to
Provincial capitals	63	34 (29 lost status)	whether its local town is still a seat
District tier	about 700 districts	abolished	distance to the nearest office
Communes	10,035	3,321	size of the lowest jurisdiction
Statistical reporting	63 provinces	34 provinces	which units post-reform data describe

The reform matters for method because its rule was published in stages before the map was final. The list of 52 provinces to be merged was released in March 2025, the pairings and the capital of each new province were announced with Resolution 60-NQ/TW on 12 April 2025, and the government reported that all 23 new provinces met the area and population standards of Resolutions 1211/2016/UBTVQH13 and 76/2025/UBTVQH15. Those standards set a population floor of 0.9 million for mountainous provinces and 1.4 million for others, together with area floors of 8,000 and 5,000 square kilometres. Supplemental Appendix S4 shows that the population floors bind on every realized group while the area floors are violated by four, so only the former can be read as a rule.

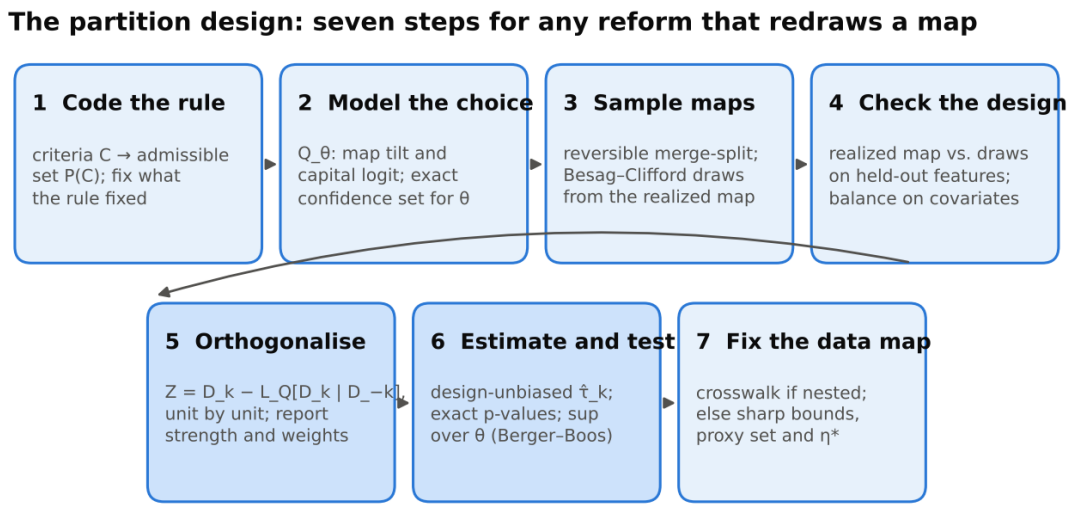
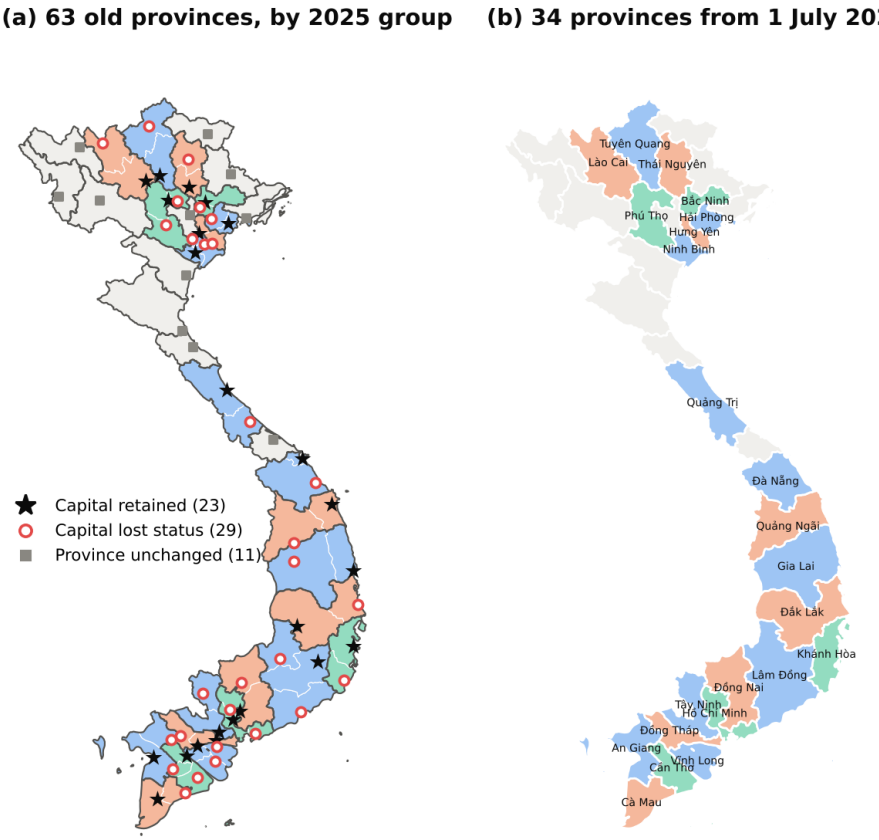


Figure 1. The partition-design protocol. Steps 1 to 4 build and check the design; steps 5 and 6 estimate and test; step 7 (a companion paper) handles data published on the new map.



Boundaries: geoBoundaries ADM1 (pre-2025), dissolved by the Resolution 202/2025/QH15 groupings. Offshore archipelagos omitted.

Figure 2. The 2025 reform. Panel (a) colours each of the 63 old provinces by the province it joined; stars mark the 23 capitals that were kept and open circles the 29 that lost their status. Panel (b) shows the 34 provinces in force from 1 July 2025. Boundaries: geoBoundaries ADM1; capital locations approximate.

The tools are not specific to Vietnam. Denmark's 2007 reform cut 271 municipalities to 98, Japan's Heisei amalgamations roughly halved the number of municipalities, China has converted hundreds of counties into urban districts, and Indonesia's district splits after 1999 run the reform in reverse. Each case shares the three features this paper studies: a bundled treatment, few reshaped groups and statistics that switch to the new map.

3. Framework

Maps and design. Let $G = (V, E)$ be the adjacency graph of the old jurisdictions and $F \subseteq V$ the free nodes that the reform may regroup. A map is a pair $P = (\pi, c)$, where π partitions F into connected groups with a given multiset of sizes and $c(g)$ selects the capital of each group from its members' capitals. The published criteria C define the admissible set $\mathcal{P}(C)$. The design is a distribution Q_θ on $\mathcal{P}(C)$ indexed by a low-dimensional θ; in our application,

$$Q_\theta(\pi, c) \;\propto\; \exp\{\lambda^\top T(\pi)\}\; \mathbf{1}\{\pi \in \mathcal{P}(C)\}\; \prod_{g \in \pi} \frac{\exp\{\gamma^\top x_{c(g)}(g)\}}{\sum_{k \in g} \exp\{\gamma^\top x_k(g)\}},$$

with map features $T(\pi)$ (such as compactness), capital covariates $x_k(g)$ and $\theta = (\lambda, \gamma)$.

Units, doses and outcomes. Units $i = 1, \dots, N$ (communes, towns or grid cells) are observed before ($t = 0$) and after ($t = 1$) the reform. A known exposure mapping $D_i(P) \in \mathbb{R}^K$ summarises what map P does to unit i: in our application L_i (the unit's capital lost status), ΔT_i (change in distance to the capital), ΔS_i (change in log population of the jurisdiction) and B_i (proximity to an old border made internal). Figure 3 maps them for Vietnam. Co-treatments that do not depend on the provincial map, such as the abolition of districts and the consolidation of communes, are collected in C_i.

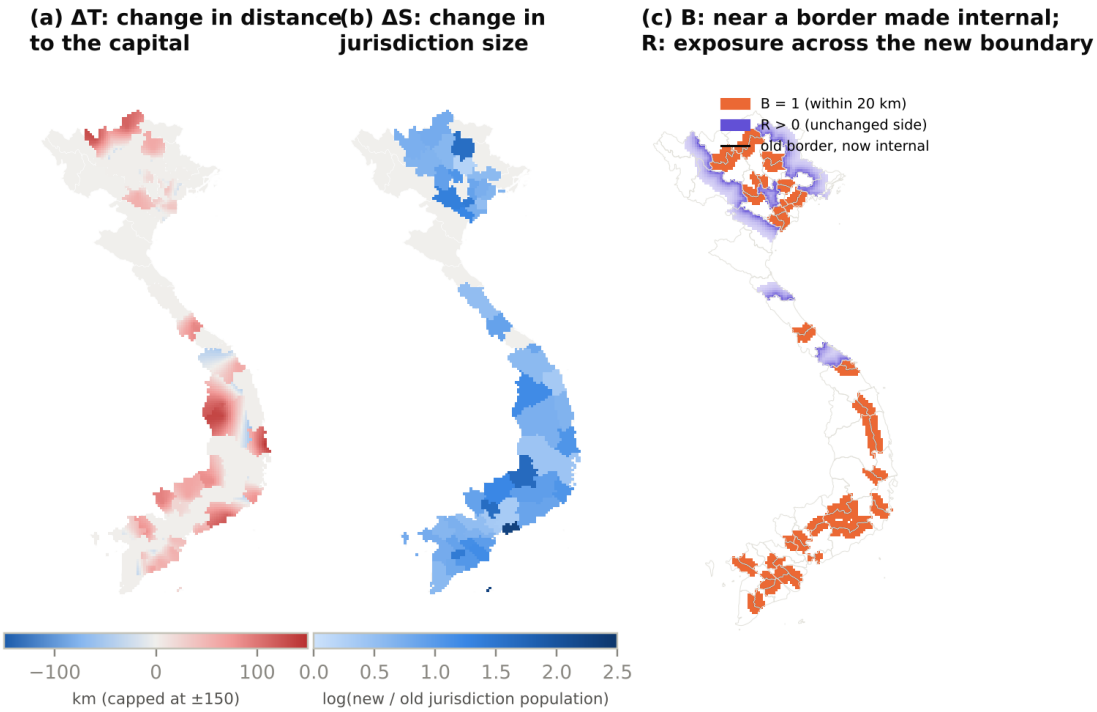


Figure 3. The dose vector on the realized map, on an 8 km grid of pseudo-communes. (a) Change in straight-line distance to the provincial capital. (b) Change in log jurisdiction population. (c) Cells within 20 km of an old border that became internal ($B = 1$) and cells in unchanged provinces next to a merged one ($R > 0$), whose exposure is fixed by the list of merged provinces and therefore not varied by the design.

Assumption A1 (exposure sufficiency). $Y_{i1}(P) = Y_{i1}\left(D_i(P); C_i\right)$ for every map P: the map reaches unit i only through its dose vector, given the co-treatments.

Assumption A2 (no anticipation). Y_{i0} does not depend on the map eventually chosen.

Assumption A3 (design). Conditional on all potential outcomes and on C, the realized map P_1 is a draw from Q_{θ_0} on $\mathcal{P}(C)$.

Assumption A4 (additive separability). $Y_{i1}\left(d; C_i\right) = Y_{i1}\left(0; C_i\right) + \sum_{k=1}^{K} f_{ki}\left(d_k\right)$ with $f_{ki}(0) = 0$, unit-specific and unrestricted otherwise.

A3 does not say the map was random. It says that once the criteria are imposed and the systematic preferences in θ are modelled, which admissible map was chosen is unrelated to units' untreated trends; Section 6 shows how to test its observable implications. A4 rules out interactions between components; any interaction of interest can be added as a further component. Write $\Delta Y_i = Y_{i1} - Y_{i0}$ and $\Delta Y_i(0) = Y_{i1}\left(0; C_i\right) - Y_{i0}$, which under A1 to A3 is fixed across admissible maps.

4. What standard estimators average

Proposition 1 (merged-by-post regressions). Let M_i indicate a merged jurisdiction, $p = N^{-1} \sum_i M_i$, and β_M the OLS slope of ΔY_i on M_i with an intercept. Under A1, A4 and linear $f_{ki}(d) = \tau_{ki} d$,

$$\beta_M = \beta_M^0 + \sum_k \sum_i \omega_{ki}\, \tau_{ki}, \qquad \omega_{ki} = \frac{(M_i - p)\, D_{ki}}{Np(1-p)},$$

where β_M^0 is the slope of $\Delta Y_i(0)$ on M_i.

The coefficient loads on every component at once. On Vietnam's realized doses the weights are 0.56 on capital loss, 0.27 on distance, 0.83 on jurisdiction size and 0.36 on internal borders (Figure 4a), and 3.8 percent of the absolute weight on distance is negative because some communes moved closer to their capital. This is an application of the multiple-treatment results of Goldsmith-Pinkham, Hull and Kolesár (2024) rather than a new phenomenon; what is specific to map changes is that the components are assigned jointly by one partition.

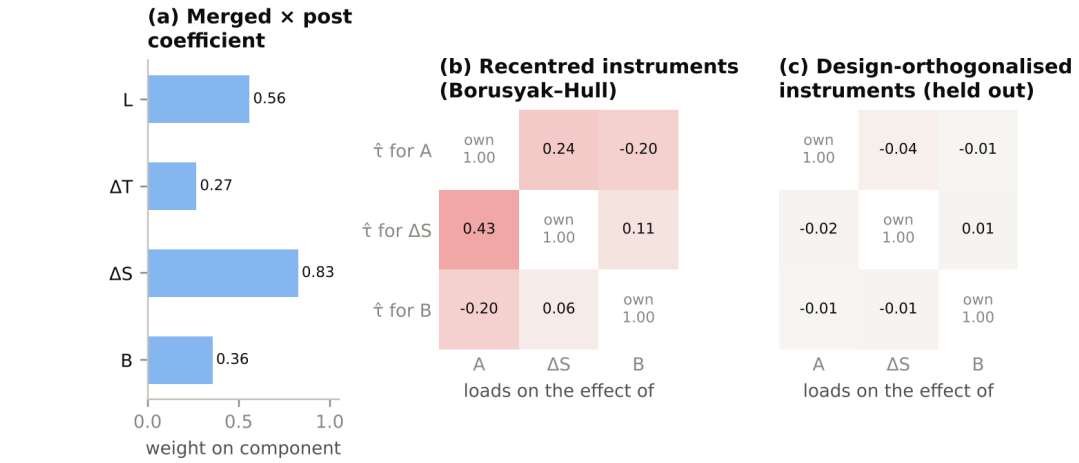
Recentring is the natural design-based response. Following Borusyak and Hull (2023), let $\mu_{ki} = E_Q[D_{ki}]$ and $Z_{ki}^r = D_{ki} - \mu_{ki}$.

Proposition 2 (contamination under recentring). Under A1 to A4 with linear f and θ known,

$$E_Q\left[\sum_i Z_{ki}^r\, \Delta Y_i\right] = \sum_i \sum_l \tau_{li}\, Cov_Q\left(D_{ki}, D_{li}\right).$$

The recentred estimator of component k therefore averages the effects of every component l with weights $Cov_Q(D_{ki}, D_{li})$, which are non-zero whenever the design moves the components together and can be negative.

Figure 4b reports these weights on Vietnam's admissible maps for the three components the design separates (Section 5): a capital-access shock A_i, equal to the change in distance to the capital and zero unless the capital was lost, jurisdiction size and internal borders. The recentred estimator of the size effect loads 0.43 on the capital-access effect, and the estimators of capital access and internal borders load 0.24 and -0.20 on the other components. Splitting capital access further into status loss and distance raises the loadings to 0.81. Recentring solves the omitted-variable problem that Borusyak and Hull (2023) target, but with a bundled treatment it does not separate the bundle.



A: capital-access shock (change in distance to the capital, zero unless the capital was lost). Each row of (b) and (c): how the estimator of one component weights the true effects of the others, relative to its own. Panel (c) fits the design projections on half of the 2,001 maps and evaluates on the other half.

Figure 4. What each estimator averages under Vietnam's design. (a) Weights of a merged-by-post coefficient on each component (Proposition 1). (b) Cross-loadings of recentred estimators (Proposition 2) for the three components the design separates: row k shows how the estimator of component k weights the true effect of each component, relative to its own. (c) The same for the design-orthogonalised estimators (Theorem 1), with the design projections fitted on half of the 2,001 maps and evaluated on the other half. 8 km grid of pseudo-communes.

5. Design-orthogonalised dose instruments

Definition (design-orthogonalised dose instruments). For unit i and component k, let $h_{ki}(\cdot)$ be either the design conditional expectation $E_Q\left[D_{ki} \mid D_{-k,i}\right]$ (version C) or the unit-specific design linear projection of D_{ki} on $\left(1, D_{-k,i}\right)$ (version L). The design-orthogonalised dose instrument (DODI) is

$$Z_{ki} = D_{ki} - h_{ki}\left(D_{-k,i}\right),$$

and the DODI estimator of component k is

$$\hat{\tau}_k^{DODI} = \frac{\sum_i Z_{ki}\,\Delta Y_i}{\sum_i E_Q\left[Z_{ki} D_{ki}\right]}.$$

Both h_{ki} and the denominator are computed from simulated maps; they depend on the design, never on outcomes.

Theorem 1 (contamination-free convex estimand). Suppose A1 to A4 hold, θ_0 is known and

$\sum_i E_Q[Z_{ki} D_{ki}] > 0$.

(a) *No contamination.* In version C, $E_Q[Z_{ki}\, f_{li}(D_{li})] = 0$ for every $l \neq k$ and every function f_{li}. In version L, the same holds whenever f_{li} is affine.

(b) *Convex weights.* In version C, if f_{ki} is absolutely continuous,

$$E_Q[Z_{ki}\, f_{ki}(D_{ki})] = E_Q\left[\int f_{ki}'(t)\, \omega_{ki}\left(t \mid D_{-k,i}\right)\, dt\right],$$
$$\omega_{ki}(t \mid \cdot) = Cov_Q\left(\mathbf{1}\{D_{ki} \geq t\}, D_{ki} \mid \cdot\right) \geq 0.$$

- In version L, if D_{ki} is binary or f_{ki} is linear with slope τ_{ki}, then $E_Q[Z_{ki} f_{ki}(D_{ki})] = \tau_{ki}\, E_Q[Z_{ki}^2]$.

(c) *Design unbiasedness.* $E_Q[\hat{\tau}_k^{DODI}] = \beta_k$, where

$$\beta_k = \frac{\sum_i E_Q[Z_{ki}\, f_{ki}(D_{ki})]}{\sum_i E_Q[Z_{ki}\, D_{ki}]}$$

- is a convex average of unit-level marginal effects of component k alone, with weights $E_Q[\omega_{ki}(t \mid \cdot)]$ over units and dose levels in version C. In version L the same conclusion holds if f_{li} is affine for every $l \neq k$ and D_{ki} is binary or f_{ki} is linear, in which case $\beta_k = \sum_i \tau_{ki} E_Q[Z_{ki}^2] \,/\, \sum_i E_Q[Z_{ki}^2]$.

Theorem 1 is the core of the framework. Part (a) is the reason to partial unit by unit under the design rather than to add the other doses as regression controls: the projection is taken over the maps a unit could have received, so it removes exactly the co-movement that the design induces for that unit. Part (b) gives the estimand a clean interpretation, as in Yitzhaki (1996) and Angrist and Imbens (1995) for a single treatment. Part (c) uses the design-expected denominator, which makes the estimator unbiased, unlike a ratio of realized sums. Theorem 1 is stated for the population design; the estimator actually computed uses the pooled set of the realized map and its S simulated companions, and possibly a sieve. Theorem 1' shows that nothing is lost in that step: the implemented estimator is exactly unbiased for a convex estimand defined on the pooled set, and its only possible contamination is an explicit sieve-approximation term.

Theorem 1' (the estimator as implemented). Let $\mathcal{M} = \{P^{(0)}, \ldots, P^{(S)}\}$ be the pooled set produced by the construction of Theorem 2(a), with $P^{(0)} = P_1$, and let $\bar{E}$ denote the average over its elements. Fix a basis $b(\cdot)$: affine for version L, all levels, squares and cross-products for the sieve version of C. Let $h_{ki}^{\mathcal{M}}$ be the unit-i least-squares projection of D_{ki} on $b\left(D_{-k,i}\right)$ over $\mathcal{M}$, $Z_{ki} = D_{ki} - h_{ki}^{\mathcal{M}}$, and

$$\hat{\tau}_k = \frac{\sum_i Z_{ki}(P_1)\, \Delta Y_i}{\sum_i \bar{E}[Z_{ki} D_{ki}]}.$$

Under A1 to A4 at θ_0, conditional on the unordered set $\mathcal{M}$:

(a) *Exact unbiasedness for a pooled-set estimand.* $E\left[\hat{\tau}_k \mid \mathcal{M}\right] = \beta_k^{\mathcal{M}} + R_k^{\mathcal{M}}$, with

$$\beta_k^{\mathcal{M}} = \frac{\sum_i \bar{E}\left[Z_{ki} f_{ki}\left(D_{ki}\right)\right]}{\sum_i \bar{E}\left[Z_{ki} D_{ki}\right]}, \qquad R_k^{\mathcal{M}} = \frac{\sum_i \sum_{l \neq k} \bar{E}\left[Z_{ki}\, r_{li}\right]}{\sum_i \bar{E}\left[Z_{ki} D_{ki}\right]},$$

- where r_{li} is the residual of $f_{li}\left(D_{li}\right)$ from its unit-i projection on $b\left(D_{-k,i}\right)$ over $\mathcal{M}$.

(b) *Explicit contamination bound.* $\left|R_k^{\mathcal{M}}\right| \leq \sum_i \bar{E}\left[Z_{ki}^2\right]^{1/2} \sum_{l \neq k} \bar{E}\left[r_{li}^2\right]^{1/2} / \sum_i \bar{E}\left[Z_{ki} D_{ki}\right]$, and $R_k^{\mathcal{M}} = 0$ whenever every f_{li}, $l \neq k$, lies in the span of b on $\mathcal{M}$: affine effects in version L, effects quadratic in the other doses in the sieve version (which covers any effect of a binary dose).

(c) *Convexity.* If $f_{ki}(d) = \tau_{ki} d$ or D_{ki} is binary, then $\beta_k^{\mathcal{M}} = \sum_i \tau_{ki} \bar{E}\left[Z_{ki}^2\right] / \sum_i \bar{E}\left[Z_{ki}^2\right]$, a convex average for every $\mathcal{M}$. For nonlinear f_{ki} the weights $\bar{E}\left[Z_{ki} \mathbf{1}\{D_{ki} \geq t\}\right]$ are computable from $\mathcal{M}$, so their negative share is a reported diagnostic rather than an assumption.

Theorem 1' replaces "unbiased up to Monte Carlo error" with an exact statement. There is no simulation bias at all: the estimator is unbiased for $\beta_k^{\mathcal{M}}$, which is itself a convex average of component k's effects, and $\beta_k^{\mathcal{M}}$ approaches the population estimand β_k at the usual $S^{-1/2}$ rate when the companions approximate Q. The engine is Lemma 2 in Appendix A: conditional on $\mathcal{M}$, the realized map is equally likely to be any of its elements. The bound in (b) is the price of a finite sieve, and it is zero for the effect shapes the basis spans. In every simulation scenario of Supplemental Appendix S2 the affine and sieve versions give mean estimates within 0.01 of each other. Orthogonality holds by construction within the pooled set, so it is not evidence by itself. Figure 4c therefore fits the projections on half of the maps and evaluates them on the other half: every held-out cross-loading is 0.04 or less in absolute value, and all unit weights are non-negative. The price is variance. We define the **design strength** of component k as $\sum_i E_Q\left[Z_{ki} D_{ki}\right] / \sum_i Var_Q\left(D_{ki}\right)$, the share of the component's design variance that survives partialling.

Design strength also reveals what the design cannot do. On every admissible map, a commune's capital is lost exactly when its distance to the capital changes, so $E_Q\left[L_i \mid \Delta T_i\right] = L_i$ and the version-C strength of capital loss given distance is zero. A version-L instrument still has strength 0.34, but only because it assumes the distance effect is linear; any curvature in that effect would load onto the capital-loss estimate. The same holds for the old capitals, whose distance to the new capital is positive exactly when they lose status. The protocol's rule is to merge such components: we use a single capital-access shock A_i for communes and capital loss L for capital towns. With $(A, \Delta S, B)$ the design strengths are 0.74, 0.71 and 0.74, and with $(L, \Delta S)$ at the capital level the strength of L is 0.87.

Proposition 3 (immunity to map-invariant co-treatments). Suppose the co-treatments C_i are the same on every admissible map, so they are fixed under Q. Then under A1 to A4 Theorem 1 holds with $f_{ki}(\cdot) =$

$f_{ki}(\cdot; C_i)$: co-treatments affect $\hat{\tau}_k^{DODI}$ only through the heterogeneity of unit effects, their own effects receive zero weight, and no untreated control group is required.

Proposition 3 answers the concern that district abolition and commune consolidation hit all 63 provinces. Those reforms change $\Delta Y_i(0)$, but the DODI estimator uses only variation across admissible provincial maps, along which they are constant. The condition fails if commune consolidation was itself shaped by the provincial grouping, for instance if new communes straddle old provincial lines in a pattern that depends on which provinces merged; in that case consolidation exposure belongs in D_i.

6. Exact inference

Sampling admissible maps. We sample from Q_θ with a merge-split kernel: pick an adjacent pair of groups uniformly, pool their nodes, and re-split the pool uniformly among all admissible splits with the same two sizes. Accept with probability

$$\min\left\{1,\ \frac{Q_\theta(\pi')\ |A(\pi)|}{Q_\theta(\pi)\ |A(\pi')|}\right\},$$

where $A(\pi)$ is the set of adjacent group pairs. Capitals are then drawn exactly from the logit.

Lemma 1 (reversibility). If the criteria are group-local (each constraint involves one group at a time), the merge-split kernel is reversible with respect to Q_θ restricted to $\mathcal{P}(C)$, and so is the joint kernel that redraws capitals from the logit after each move. Because every re-split keeps the two sizes of the pair it replaces, the global multiset of group sizes is preserved.

Theorem 2 (finite-sample validity). Consider the fully specified null $H_0(\tau^0)$: $\Delta Y_i = \Delta Y_i(0) + \sum_k f_{ki}^0(D_{ki})$ for known f^0. Let $\Delta\widetilde{Y}_i = \Delta Y_i - \sum_k f_{ki}^0(D_{ki}(P_1))$ and let $T\left(P, \Delta\widetilde{Y}\right)$ be any statistic.

(a) *Exactness without mixing.* Run any Q_θ-reversible kernel m steps from P_1 to a hub P_*, and S independent m-step runs from P_* to $P^{(1)}, \ldots, P^{(S)}$. Under A1 to A3 with $\theta = \theta_0$, the p-value

$$\hat{p}(\tau^0; \theta) = \frac{1 + \sum_s \mathbf{1}\{T\left(P^{(s)}, \Delta\widetilde{Y}\right) \geq T\left(P_1, \Delta\widetilde{Y}\right)\}}{S+1}$$

- satisfies $\Pr(\hat{p} \leq \alpha) \leq \alpha$ for every α, for every $m \geq 1$ and $S \geq 1$.

(b) *Estimated design.* Let Θ_γ be any set, possibly computed from the realized map, with $\Pr\left(\theta_0 \in \Theta_\gamma\right) \geq 1 - \gamma$. Then $\hat{p}_{BB} = \sup_{\theta \in \Theta_\gamma} \hat{p}(\tau^0; \theta) + \gamma$ satisfies $\Pr(\hat{p}_{BB} \leq \alpha) \leq \alpha$.

(c) *Composite nulls.* Suppose the other components have constant effects, $f_{li}(d) = \tau_l d$ for $l \neq k$. For

$H_0 : \tau_k = t$, the same holds with the supremum taken over a joint $1-\gamma$ confidence set for (θ, τ_{-k}), or over the product of separate sets at levels $1-\gamma_1$ and $1-\gamma_2$ with $\gamma = \gamma_1 + \gamma_2$ added.

Part (a) is the Besag and Clifford (1989) parallel construction applied to Lemma 1: exchangeability of the realized and simulated maps does not depend on how well the chain mixes, which affects only power (Chikina, Frieze and Pegden 2017; Howes 2023). Part (b) answers the objection that fitting θ to the realized map breaks exchangeability. The bound needs no independence between Θ_γ and the test, because it is a union bound over the event that θ_0 escapes the set (Berger and Boos 1994; Dufour 2006). We build Θ_γ by inverting Monte Carlo tests of θ on the realized capital choices and map features. For a candidate $\theta^0 = (\lambda^0, \gamma^0)$ the test runs the Besag–Clifford construction at λ^0 from the realized map, draws capitals from the logit at γ^0 on every companion (the realized map keeps its realized capitals), and ranks the Mahalanobis distance of $\left(T(\pi), \sum_g x_{c(g)}(g)\right)$ from its pooled mean, with the pooled covariance. Each such test is exact by part (a), so $\Theta_\gamma = \{\theta^0 : \hat{p}\,(\theta^0) > \gamma\}$ covers θ_0 with probability at least $1-\gamma$ on the grid that defines the design. Proposition 4 below shows that reweighting one set of draws to another λ is also exact, not an approximation, which makes fine grids cheap; it costs only effective sample size. For the test statistic itself, the design quantities h_{ki} and denominators must be symmetric functions of the pooled set of maps, as in Section 5, so that the realized map and its companions are treated identically.

Theorem 2(c') (heterogeneous nuisance effects). Consider the weak null H_{0k}: $f_{ki} \equiv 0$ for every i, with the other effects f_{li} heterogeneous across units and in the span of b. Hold fixed $\Delta\ddot{Y} = \Delta Y - \sum_{l \neq k} \hat{\tau}_l D_l\,(P_1)$ and use the studentised statistic $t(P) = \sum_i Z_{ki}(P)\Delta\ddot{Y}_i / \widehat{V}(P)^{1/2}$, with $\widehat{V}(P) = \sum_c \left(\sum_{i \in c} Z_{ki}(P)\Delta\ddot{Y}_i\right)^2$ summed over the blocks c of Theorem 3. Under the conditions of Theorem 3, and with companions that approximate draws from Q_{θ_0}, the randomization test that ranks $t\,(P_1)$ among $t\left(P^{(0)}\right), \ldots, t\left(P^{(S)}\right)$ has asymptotic size at most α.

Part (c') removes the homogeneity that part (c) needs, at the cost of exactness. The argument (Appendix A) has three steps: Theorem 1'(a) gives the numerator mean zero under H_{0k} whatever the other effects are; $\widehat{V}$ overestimates its variance, because it also picks up the squared block means; and the reference distribution of t is asymptotically standard normal, as in the studentised permutation tests of Chung and Romano (2013) and Wu and Ding (2021). The Monte Carlo of Section 7.5 runs the test with heterogeneous nuisance effects, an independent truth and a short analyst chain.

Proposition 4 (exact tests under a reweighted design). Generate companions with a Q-reversible kernel as in Theorem 2(a), but suppose the realized map is drawn from Q^*, with $w(P) = Q^*(P)/Q(P)$ known up to a constant. Then, conditional on $\mathcal{M}$, $\Pr\left(P_1 = P^{(s)} \mid \mathcal{M}\right) = w\left(P^{(s)}\right) / \sum_r w\left(P^{(r)}\right)$, and the weighted p-value $\hat{p}_w = \sum_s w\left(P^{(s)}\right) \mathbf{1}\{T\left(P^{(s)}\right) \geq T\,(P_1)\} / \sum_s w\left(P^{(s)}\right)$, with both sums over all $S+1$ elements, satisfies $\Pr\left(\hat{p}_w \leq \alpha\right) \leq \alpha$.

Proposition 4 has two uses. It turns one chain into exact tests at every value of a tilt, which we use to run Vietnam's placebo over the whole unbounded confidence set for λ (Section 8). And it gives an exact sensitivity analysis for A3.

Sensitivity to departures from A3. Let u be an unobserved unit-level trend and $z_k(P;u)$ the alignment $\sum_i D_{ki}(P)u_i$, standardised over $\mathcal{M}$. The departure

$$Q^*_\Gamma(P) \;\propto\; Q_\theta(P)\,\exp\{\Gamma\, z_k(P;u)\}$$

makes maps that give high doses to high-trend units e^Γ times more likely per standard deviation of alignment; $\Gamma = 0$ is A3. Proposition 4 gives the exact p-value for every (Γ, u). We report two numbers for each conclusion, in the spirit of Rosenbaum (2002), Oster (2019) and Cinelli and Hazlett (2020). The breakdown Γ^* is the smallest $|\Gamma|$ at which the conclusion changes when u is the outcome itself, the worst case for an unobserved trend. The benchmark $\hat{\Gamma}_{pre}$ is the tilt at which the realized map's alignment with pre-reform growth is typical, that is, how strongly the actual map selected on the trend we can see. A conclusion is robust when Γ^* exceeds $\hat{\Gamma}_{pre}$: an unobserved trend would have to steer the map more strongly than the observed one did.

Theorem 3 (large-sample approximation). Consider a sequence of reforms, or of pooled reforms, in which the free graph splits into $G_n \to \infty$ blocks that the design treats independently, with bounded block size, bounded moments of $Z_{ki}\Delta Y_i$ and $G_n^{-1}Var_Q\left(\sum_i Z_{ki}\Delta Y_i\right)$ bounded away from zero. Then $\left(\hat{\tau}_k^{DODI} - \beta_k\right)/\sigma_n \Rightarrow N(0,1)$, where $\sigma_n^2 = Var_Q\left(\sum_i Z_{ki}\Delta Y_i\right) / \left(\sum_i E_Q\left[Z_{ki}D_{ki}\right]\right)^2$. The variance under the sharp null is computable from the simulated maps, and under heterogeneity it is conservatively estimable.

Theorem 3 is a Lyapunov central limit theorem over independent blocks; extensions to weak dependence between neighbouring blocks follow from dependency-graph bounds proved by Stein's method (Chen and Shao 2004; Ross 2011). It explains when Wald intervals are acceptable, namely many independently regrouped regions or pooled reforms. A single national map such as Vietnam's is not such a case, which is why we rely on Theorem 2 and use Theorem 3 only for power calculations and planning.

Checking the design. A3 has testable implications. Under the null that the realized map is a draw from Q_θ, any feature of the map that the analyst did not use to fit θ has an exact Besag–Clifford p-value, and so does the realized imbalance of any pre-reform covariate between units with different doses. Both checks have known limits. One map is one draw, so the feature check has limited power against mild misspecification, and a covariate used to fit θ cannot test the fit; only held-out features and covariates are informative. Supplemental

Appendix S4 shows, however, that the covariate balance check is powerful against the misspecification that matters most in our Monte Carlo, an ignored capital-choice rule.

7. Application I: Denmark's 2007 municipal reform with parish micro-data

Vietnam's reform is too recent for outcome data below the province, so we first run the full protocol on a reform with a published rule, a decade and a half of post-reform outcomes and open micro-data. Denmark's Structural Reform, agreed in June 2004 and in force from 1 January 2007, cut the number of municipalities from 271 to 98. The agreement asked every new municipality to reach roughly 20,000 inhabitants, left local councils to choose their partners within that constraint, and placed each new town hall in one of the merging municipalities (Lassen and Serritzlew 2011; Blom-Hansen, Houlberg, Serritzlew and Treisman 2016). That is a partition-change design in the sense of Section 3: a floor, contiguity and a host choice define the admissible maps, and the discretion left to councils is what Q_θ must model. Figure 5 shows the reform as we rebuild it.

7.1 Data and the coded rule

Every input is public and was retrieved programmatically. Annual population for each Danish parish, 1996 to 2022, comes from Statistics Denmark table SOGN10 through the StatBank API; the table is published on a fixed parish geography, so units are the same before and after the reform. Parish polygons come from the DAGI boundary files in the `dagirlite` package on CRAN. The official 2006 key from parishes to pre-reform municipalities is table BEF4B, which also supplies each old municipality's official 2006 population. Parishes merged after 2006 carry new codes (Ål, Ho and Oksby became Blåvandshuk parish, for example), so we split their names into the 2006 component parishes and assign the old municipality that held most of the components' 2006 population; 51 parishes are matched this way and 78, all inside a single new municipality, by their nearest matched neighbour. Former town halls are the old municipalities' seat towns, geocoded with OpenStreetMap and accepted only inside the municipality (208 of 269); the remaining 61 fall back to the most populous parish. The crosswalk yields 269 old municipalities: 237 merged into 65 new ones (12 pairs, 22 triples, 16 groups of four, 9 of five, 4 of six and 2 of seven) and 32 left unchanged. We keep the 1,637 parishes in merged municipalities with positive population in every year.

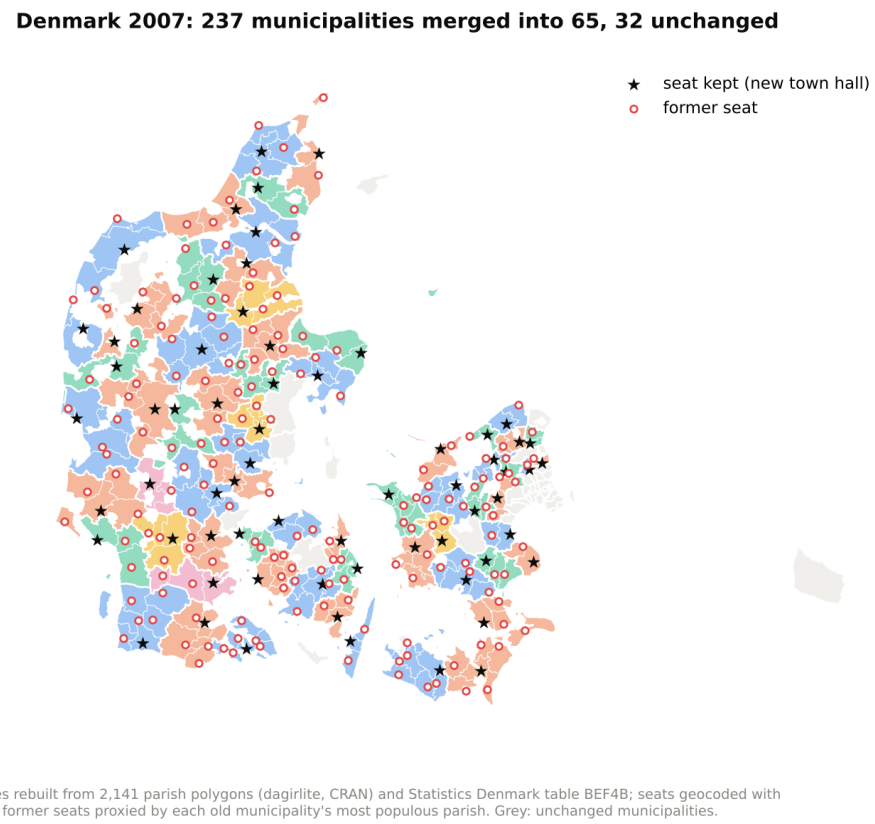


Figure 5. Denmark's 2007 reform rebuilt from open data. Each colour is one new municipality formed by a merger; grey municipalities were unchanged. Stars mark the old municipality that hosts the new town hall, open circles the former seats that lost their status.

We take the 237 merging municipalities as given and define the admissible set as all partitions of them into contiguous groups with the realized multiset of group sizes and an official 2006 population at or above a floor. One realized group falls below 20,000, at 14,120, so the floor was a rule with exceptions; we impose 14,120, the strictest floor the realized map satisfies. Contiguity uses polygons buffered by 3 km. The host is a conditional logit on log population and population-weighted centrality, with coefficients 3.22 (s.e. 0.73) and 2.02 (s.e. 0.94); it picks the actual host in 86 percent of the 65 groups.

7.2 Checking the design

The uniform design fails. Realized seats within a merged group lie 15.6 km apart on average, against 20.9 km for the average admissible map (exact p = 0.002), and the realized groups are more equal in population (p = 0.012; Table 2). Councils chose compact, balanced partners, which is exactly the discretion Q_θ exists to model. We therefore tilt the design by $\exp\left(-\lambda \sum_g \bar{d}_g\right)$, where $\bar{d}_g$ is the mean distance between the seats of group g. At $\lambda = 0.4$ the realized map is central on the fitted feature (p = 0.98) and typical on the three

held-out ones (p = 0.074, 0.51 and 0.47).

Table 2. Denmark design check. The realized 2007 map against 1,000 Besag–Clifford companions under the uniform design and under the compactness-tilted design at $\lambda = 0.4$; exact two-sided rank p-values, bold below 0.05.

Feature	Realized	Uniform: mean	Uniform: p	Tilted: mean	Tilted: p
Mean distance between merged seats (km), fitted	15.59	20.94	**0.002**	15.60	0.979
Coefficient of variation of group population	0.524	0.606	**0.012**	0.553	0.074
Median group population (thousand)	45.58	43.98	0.360	43.88	0.513
Host is the most populous member (fitted rule)	0.819	0.786	0.082	0.813	0.474

The joint 99 percent set for $\theta = (\lambda, \gamma)$ is built as Theorem 2(b) requires: on a grid of 9 values of λ from 0.15 to 0.6 and 81 host-logit values within ± 3 standard errors, we run a separate Besag–Clifford chain of 400 companions at each λ, draw hosts at each γ, and invert the exact Mahalanobis test of Section 6. The set keeps 236 of the 729 grid points (Figure 7a). It spans λ from 0.25 to the top of the grid and almost the whole host-coefficient box, so one map pins the design down only loosely. That matters for how the Berger–Boos column below should be read.

7.3 Doses, outcomes and estimators

The components are those the Vietnam analysis separates: the capital-access shock A_i, the change in a parish's distance to its seat in units of 10 km, zero unless its old municipality lost the seat; the jurisdiction-size dose ΔS_i, the log ratio of the new to the old municipality's population; and B_i, an indicator for lying within 3 km of an old border that became internal. Their design strengths are 0.69, 0.58 and 0.68. The outcomes are trend breaks in annual log population growth in percentage points: the 2006–16 average minus the 2000–06 average, and the same over 2006–22. Two placebos use only pre-reform data: growth over 2000–06, and the trend break 2003–06 against 2000–03.

We compare the design-based estimator with five estimators applied researchers use: a multi-dose regression clustered by new municipality; difference-in-differences with the 337 parishes of the 32 unchanged municipalities as controls; a triple difference that adds a merged-municipality shift; instrumental variables with the design-expected dose $\mu_i = E_Q[D_i]$ as instrument, the shift-share analogue (first-stage R^2 of 0.52, 0.90 and 0.76); and synthetic difference-in-differences (Arkhangelsky, Athey, Hirshberg, Imbens and Wager 2021) for seat loss at the municipality level, with 31 unchanged municipalities as donors and 171 seat-losing municipalities treated. The design-based estimate is the Hodges–Lehmann centre of the exact 95 percent set,

the value at which the realized statistic sits at the median of its reference distribution, with the plug-in exact p-value and the studentised p-value of Theorem 2(c').

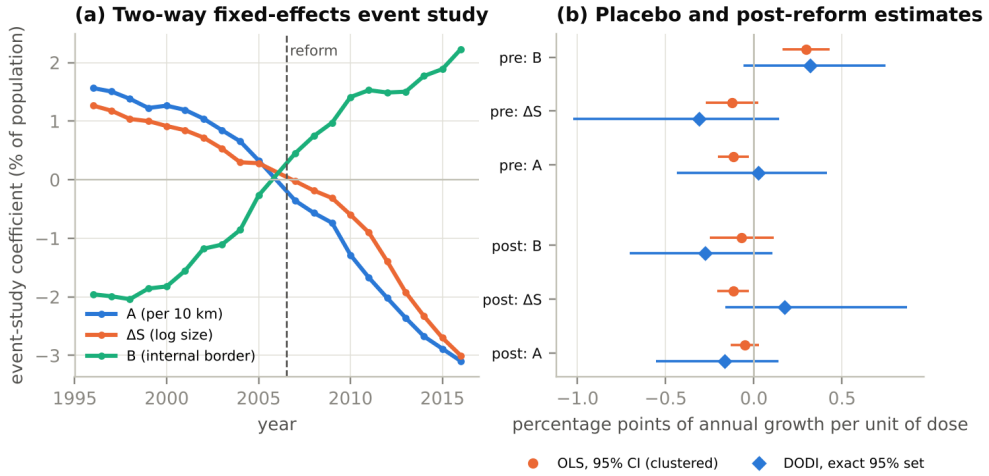
Table 3. Denmark 2007, parish micro-data (N = 1,637): standard estimators. Percentage points of annual population growth per unit of dose (A: 10 km; ΔS: log points; B: indicator). Pre growth: growth 2000–06 (placebo); pre break: 2003–06 minus 2000–03 (placebo); post: growth after 2006 minus 2000–06. OLS: all three doses, clustered by new municipality, with its p-value; DiD: unchanged municipalities as controls; DDD: adds a merged-municipality indicator; IV-μ: doses instrumented by their design expectation. Bold marks p ≤ 0.05.

Outcome, dose	OLS	p cl.	p DiD	p DDD	p IV-μ
Pre growth, A	−0.115	**0.012**	**0.011**	**0.011**	**0.004**
Pre growth, ΔS	−0.124	0.112	0.057	0.110	0.338
Pre growth, B	0.297	**<.001**	**<.001**	**<.001**	**<.001**
Pre break, A	−0.137	0.127	0.130	0.125	**0.006**
Pre break, ΔS	0.009	0.922	0.811	0.922	0.159
Pre break, B	0.087	0.479	0.339	0.477	0.932
Post 2006–16, A	−0.051	0.221	0.108	0.218	0.266
Post 2006–16, ΔS	−0.117	**0.013**	**0.005**	**0.012**	**0.001**
Post 2006–16, B	−0.069	0.457	0.092	0.455	0.913
Post 2006–22, A	−0.016	0.688	0.468	0.687	0.067
Post 2006–22, ΔS	−0.173	**<.001**	**<.001**	**<.001**	**<.001**
Post 2006–22, B	−0.093	0.283	0.059	0.281	0.893

Table 4. Denmark 2007: design-based estimates for the same outcomes. DODI: Hodges–Lehmann centre of the exact 95 percent set (test inversion); p: plug-in exact p-value; p stud.: studentised p-value of Theorem 2(c'). Evaluated at $\lambda = 0.4$ and the fitted host rule with 400 companions. Bold marks p ≤ 0.05.

Outcome, dose	DODI	Exact 95% set	p	p stud.
Pre growth, A	0.025	[−0.44, 0.42]	0.888	0.863
Pre growth, ΔS	−0.310	[−1.02, 0.15]	0.165	0.135
Pre growth, B	0.320	[−0.06, 0.75]	0.090	0.249
Pre break, A	0.015	[−0.42, 0.81]	0.958	0.933
Pre break, ΔS	−0.080	[−0.73, 0.52]	0.793	0.808

Outcome, dose	DODI	Exact 95% set	p	p stud.
Pre break, B	0.140	[−0.38, 0.75]	0.539	0.833
Post 2006–16, A	−0.165	[−0.56, 0.14]	0.264	0.110
Post 2006–16, ΔS	0.175	[−0.16, 0.87]	0.249	0.145
Post 2006–16, B	−0.275	[−0.71, 0.10]	0.160	0.339
Post 2006–22, A	−0.175	[−0.53, 0.10]	0.204	0.060
Post 2006–22, ΔS	0.160	[−0.17, 0.81]	0.389	0.309
Post 2006–22, B	−0.235	[−0.58, 0.18]	0.214	0.439



Parish population, Statistics Denmark SOGN10. (a) log population on dose × year, parish and year fixed effects, 2006 omitted. (b) pre: growth 2000–06; post: change in annual growth, 2006–16 relative to 2000–06. DODI sets are exact map-randomization sets from the fitted design (compactness tilt and host logit).

Figure 6. Denmark: why the comparison matters. (a) Two-way fixed-effects event study of log parish population on each dose interacted with year, 2006 omitted: every dose shows a trend that starts a decade before the reform. (b) Placebo (pre) and outcome (post) estimates: OLS with 95 percent clustered intervals and the design-orthogonalised estimator with exact 95 percent sets.

7.4 Results

The standard estimators fail their own checks. Every regression-based estimator finds a significant "effect" of capital access and of internal borders on growth before the reform, with p-values between below 0.001 and 0.011; instrumenting with the expected dose makes the capital-access placebo worse, not better (p = 0.004 on pre-reform growth and 0.006 on the pre-reform break), because the expected dose is itself a function of geography. The event study in Figure 6a shows smooth divergence from 1996 onward for every dose, the pattern parallel-trends diagnostics are meant to catch (Rambachan and Roth 2023). The same estimators then report a significant cost of jurisdiction size: 0.12 percentage points of annual growth per log point of size

over 2006–16 and 0.17 over 2006–22, with p-values from below 0.001 to 0.013. Rural parishes far from the larger towns were already declining before 2007, and the merger rule sent exactly those parishes into large municipalities with distant seats.

The design-based tests do not. At the plausible design, none of the six design-based placebo tests rejects, plug-in or studentised, and the smallest placebo p-value is 0.090 (internal borders on pre-reform growth). None of the outcome tests rejects either. Synthetic difference-in-differences for seat loss also passes its placebo (an estimate of -0.07 percent with the reform moved to 2002, p = 0.93) and finds no significant effect (-1.5 percent, p = 0.33).

What the design can and cannot learn. A non-rejection is informative only if the test could have rejected. Table 5 therefore reports the minimum detectable effect at 80 percent power, computed from the same companions by treating each in turn as the realized map. For capital access it is 0.26 percentage points per 10 km, five times the regression estimate; for internal borders 0.67. For jurisdiction size the design cannot reach 80 percent power below 2 percentage points per log point, more than ten times the regression estimate. This is the substantive lesson of the Danish case. Conditional on the capital-access and border doses and on the size multiset that the agreement fixed, the reform left almost no variation in jurisdiction size that is free of geography. The significant size effects that regressions report therefore rest entirely on a parallel-trends assumption the placebos reject, and this reform cannot support a credible estimate of the cost or benefit of larger municipalities for population growth. A rule-based design says so; a regression does not.

Table 5. Denmark robustness for the placebo on pre-reform growth and the two outcomes. p, min over set: the smallest plug-in p-value over the 236 points of the joint 99 percent design set. p BB: Berger–Boos p-value (supremum plus 0.01). MDE: minimum detectable effect at 80 percent power, in the units of Tables 3 and 4. Γ*: the smallest selection strength on an unobserved trend aligned with the outcome that changes the conclusion. Γ pre: the selection strength on observed pre-reform growth that makes the realized map's alignment typical.

Outcome, dose	p, min over set	p BB	MDE (80%)	Abs. OLS	Γ*	Γ pre
Pre-reform growth, A	0.219	1.00	0.34	0.12	2.0	0.3
Pre-reform growth, ΔS	0.090	1.00	> 2	0.12	0.7	−0.9
Pre-reform growth, B	0.015	1.00	0.71	0.30	0.4	1.0
Outcome 2006–16, A	0.020	0.92	0.26	0.05	1.1	0.3
Outcome 2006–16, ΔS	0.045	0.89	> 2	0.12	1.1	−0.9
Outcome 2006–16, B	0.045	0.40	0.67	0.07	0.8	1.0

Outcome, dose	p, min over set	p BB	MDE (80%)	Abs. OLS	Γ*	Γ pre
Outcome 2006–22, A	0.015	0.69	0.26	0.02	1.2	0.3
Outcome 2006–22, ΔS	0.090	0.99	> 2	0.17	1.4	−0.9
Outcome 2006–22, B	0.035	0.67	0.59	0.09	1.0	1.0

Robustness and sensitivity. Berger–Boos p-values over the joint set are at least 0.40 for every test (Table 5), so no conclusion depends on the point at which the design is evaluated. That robustness is cheap, however, because the set is wide: the smallest p-values within it fall to 0.015–0.045, so at some admissible design parameters the outcome tests for capital access and size would reject and internal borders would fail the placebo. The Berger–Boos correction exists so that no such point drives a conclusion, and with a single map it leaves the Danish results where the plug-in tests put them: no effect that survives the design's own uncertainty. The sensitivity analysis of Section 6 separates the components more sharply (Figure 7b). For capital access the realized map's selection on observed pre-reform growth is weak ($\hat{\Gamma}_{pre} = 0.3$) and a conclusion would need selection of $|\Gamma^*| = 1.1$ on an outcome-aligned trend to change, nearly four times as strong. For jurisdiction size the observed selection is -0.9 and the breakdown 1.1. For internal borders the observed selection, 1.0, already exceeds the breakdown of 0.8. Councils dissolved borders next to parishes that were growing faster than the borders that other admissible partnerships would have dissolved, so A3 is doubtful for B, as the regressions' placebo failure and the minimum p-value in the set also indicate, and we do not interpret its estimates. This is the diagnosis the framework is meant to deliver before any outcome is examined.

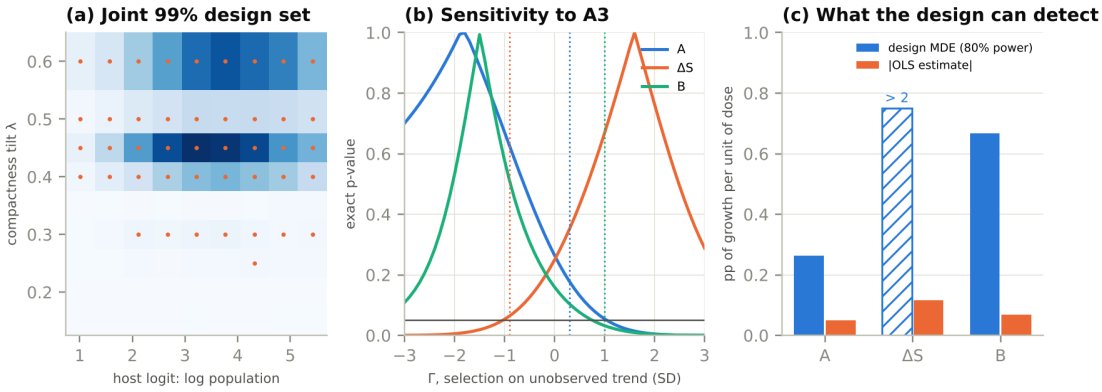


Figure 7. Denmark robustness. (a) The joint 99 percent set for the compactness tilt and the host-logit coefficient on log population (dots; shading is the exact p-value, maximised over the centrality coefficient). (b) Exact p-values for the 2006–16 outcome under the A3 departure $Q^*_\Gamma \propto Q_\theta e^{\Gamma z}$ with an outcome-aligned trend; dotted lines mark $\hat{\Gamma}_{pre}$, the selection on observed pre-reform growth. (c) Minimum detectable effects

at 80 percent power against the absolute regression estimates.

7.5 A Monte Carlo with an independent truth

The commune-grid simulations of Supplemental Appendix S2 draw the true map from the same exchangeable pool the analyst uses, so they cannot reveal chain or implementation error. Here the truth is generated separately. A long merge-split chain at $\lambda = 0.35$, run for 20,000 burn-in steps and then sampled every 2,000 steps, supplies 200 true Danish maps, each with hosts drawn from the fitted logit. For each true map the analyst runs only a short Besag–Clifford chain, 200 companions of 20 steps each, about one hundredth of the truth chain's spacing. Outcomes combine a map-invariant geographic trend (rural decline with distance to the eight largest towns and a north–south gradient), effects of the non-null components that vary across old municipalities with standard deviation 0.3 around a mean of 0.3, and noise.

Table 6. Monte Carlo on Danish geography with an independent truth: rejection rates of a true null at 5 percent and mean estimates of the null component (truth zero), 200 replications. The Monte Carlo standard error of a 5 percent rejection rate is 0.015. Bold marks rates outside two standard errors of 0.05 and biases above 0.1.

Scenario	OLS	Recentred	DODI, plug-in	DODI, studentised	Bias: recentred	Bias: DODI
All effects zero, geographic trend	**0.130**	0.015	0.030	0.030	−0.001	−0.005
A null; ΔS and B heterogeneous	0.075	0.030	0.030	0.040	0.009	−0.010
ΔS null; A and B heterogeneous	0.030	0.075	0.055	0.065	**0.324**	0.003

Three results matter (Table 6). The regression over-rejects when untreated trends follow geography, at 13 percent. Recentring is contaminated exactly as Proposition 2 predicts once the other components have effects: its estimate of a size effect that is truly zero averages 0.32, close to the 0.3 mean effects of the other components it loads on. The design-orthogonalised estimator stays within 0.01 of zero in every scenario, and both of its tests reject between 3.0 and 6.5 percent of the time, inside the Monte Carlo band, even though the analyst's chain is short and was not the process that generated the map, and the nuisance effects are heterogeneous. That is what Theorem 2(a) promises for exactness without mixing and what Theorem 2(c') promises for heterogeneity.

8. Evidence from Vietnam's 2025 consolidation

Vietnam's reform, which cut 63 provinces to 34 under published population and area floors, is too recent for outcomes below the province, so we use it for a placebo and for simulations on its real geography. The details are in Supplemental Appendices S1 to S4; this section reports what matters for the argument.

A placebo on the real map. The reform cannot have changed characteristics fixed before it was announced, so we treat five of them as outcomes for each cell of an 8 km grid of 5,153 commune-sized units: the 2023 population density and population of its old province, its latitude, its distance to Hanoi or Ho Chi Minh City, and its distance to its old provincial capital. Every rejection is a false positive. Across the 15 combinations of characteristic and component (Supplemental Table S1), the multi-dose regression rejects 5 times with clustered errors and 4 times with Conley errors; recentred and design-orthogonalised tests reject none.

The regressions fail in a telling pattern. The jurisdiction-size dose "affects" latitude with $p < 0.001$ because merged provinces in the south were larger, and the capital-access dose "affects" distance to the old capital with $p < 0.001$ because remote communes are the ones that move further from a seat. Spatial standard errors do not help (Conley 1999), because the problem is the correlation of the doses with the geography that the rule itself produces, not the correlation of errors. Every design-based test passes, under the Berger–Boos version over the capital rule, under two recodings of the admissible set, and, by Proposition 4, at every value of the compactness tilt from -60 to 120, where the smallest p-value is 0.20.

Planted effects and test size. On the same grid we plant effects and confound untreated changes with the design and with a national co-treatment (Supplemental Appendix S2). With no effects at all, the multi-dose regression rejects the capital-access null in 45 percent of samples. Recentring removes the confounding but not the bundle: with homogeneous effects and no confounding it estimates a size effect of 0.40 at 0.17. The design-orthogonalised estimator stays within 0.03 of its estimand in every scenario, at a cost in power (0.46 against 0.93 for the regression when the regression is valid). At the capital level, with only about 20 effective groups, clustered OLS and the wild cluster bootstrap over-reject, at up to 20 percent, when untreated changes are spatially correlated or tied to the traits that decide which capital is kept, while the design-based tests hold their size whenever the design is right (Supplemental Appendix S3).

The design itself. The realized 2025 map lies between the 59th and 90th percentiles of 2,000 admissible maps on seven features, and a conditional logit for the choice of capital predicts 78 percent of actual choices (Supplemental Appendix S4). Vietnam therefore passes both checks of Section 6 once the capital rule is modelled.

9. Conclusion

A redrawn map is a treatment with structure, and a partition design uses that structure rather than ignoring it. The published rule supplies a design, so inference no longer depends on how many groups the reform happened to create. On Vietnam's real map, regressions find reform effects on characteristics the reform could not have changed; partition-design tests do not. On Denmark's 2007 reform, four standard estimators report a significant cost of larger jurisdictions and fail their pre-reform placebos; the design passes its placebos and shows that the reform could not have revealed such a cost. Unit-level design projections separate the components of the bundle that the design can separate, so each estimate is a convex average of one component's effects, and design strength flags the components it cannot. National co-treatments that the map does not move fall out, and reweighting the simulated maps turns every question about the design assumption into an exact test.

The approach also has a clear failure mode, and the protocol puts its diagnosis first. If the realized map sits in the tail of the admissible maps, or if pre-reform covariates are imbalanced in ways the fitted rule does not predict, the design assumption is doubtful. Denmark's reform fails the first check under a uniform design and passes once councils' preference for compact partners is modelled; the sensitivity analysis then shows that councils selected on trends near the borders they dissolved, which the protocol reports rather than hides. The framework is equally explicit about power: a design-based null comes with the effect it could have detected. Vietnam's 2025 reform passes both checks once the capital rule is modelled, and the framework flags the one separation (status loss versus distance) that the design cannot deliver.

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Appendix A. Proofs

Proof of Proposition 1. With a binary regressor and an intercept, $\beta_M = \sum_i (M_i - p)\,\Delta Y_i / \{Np(1-p)\}$. Substituting $\Delta Y_i = \Delta Y_i(0) + \sum_k \tau_{ki} D_{ki}$ gives the result. ▫

Proof of Proposition 2. $E_Q[Z^r_{ki}] = 0$ and $\Delta Y_i(0)$ is fixed under Q, so $E_Q[Z^r_{ki}\Delta Y_i(0)] = 0$. For the treatment part, $E_Q[Z^r_{ki}\tau_{li}D_{li}] = \tau_{li} Cov_Q(D_{ki}, D_{li})$. Summing over i and l gives the result. ▫

Proof of Theorem 1. Fix i and k, write $X = D_{-k,i}$ and $D = D_{ki}$. Under A1 to A3, $\Delta Y_i(0)$ is fixed under Q and $E_Q[Z_{ki}] = 0$ in both versions (by iterated expectations in version C, and because the projection includes an intercept in version L). Hence $E_Q[Z_{ki}\Delta Y_i] = \sum_l E_Q[Z_{ki} f_{li}(D_{li})]$ by A4.

(a) For $l \neq k$, $f_{li}(D_{li})$ is a function of X. In version C, $E_Q\left[\left(D - E_Q[D \mid X]\right) g(X)\right] = 0$ for every integrable g by iterated expectations. In version L, the projection residual is orthogonal to every affine function of X.

(b) In version C, $E_Q[Z_{ki} f_{ki}(D) \mid X] = Cov_Q(D, f_{ki}(D) \mid X)$. For absolutely continuous f and a reference point d_0 at or below the support, $f(D) = f(d_0) + \int_{d_0}^{\infty} f'(t)\mathbf{1}\{D \geq t\}\,dt$. Hence $Cov(D, f(D) \mid X) = \int f'(t)\, Cov(D, \mathbf{1}\{D \geq t\} \mid X)\, dt$ by Fubini, and

$Cov\left(D, \mathbf{1}\{D \geq t\} \mid X\right) = \Pr(D \geq t \mid X)\{E[D \mid D \geq t, X] - E[D \mid X]\} \geq 0$. In version L with $f(D) = \tau D$ (which covers binary D, where any f with $f(0) = 0$ is linear), $E_Q[ZD] = E_Q\left[Z\left(Z + h(X)\right)\right] = E_Q\left[Z^2\right]$, because Z is orthogonal to the affine function $h(X)$.

(c) The denominator is a design constant, so $E_Q\left[\hat{\tau}_k\right] = \sum_i E_Q\left[Z_{ki}\Delta Y_i\right] / \sum_i E_Q\left[Z_{ki}D_{ki}\right]$, and by (a) the numerator equals $\sum_i E_Q\left[Z_{ki} f_{ki}\left(D_{ki}\right)\right]$; in version L this step needs f_{li} affine for all $l \neq k$. The weights in (b) are non-negative, and they sum to the denominator when $f'_{ki} \equiv 1$ (version C), or equal $E_Q\left[Z_{ki}^2\right]$ with $\sum_i E_Q\left[Z_{ki}D_{ki}\right] = \sum_i E_Q\left[Z_{ki}^2\right]$ (version L), which gives convexity. ▫

Proof of Proposition 3. If C_i is constant on $\mathcal{P}(C)$, then $Y_{i1}\left(0; C_i\right)$ and the functions $f_{ki}\left(\cdot\,; C_i\right)$ are fixed under Q, and the proof of Theorem 1 applies verbatim. Effects of C_i enter only $\Delta Y_i(0)$, which has zero design covariance with Z_{ki}. No step uses an untreated unit. ▫

Proof of Lemma 1. Let π' be obtained from π by merging and re-splitting the adjacent pair (g, h) with pooled node set U. The forward proposal probability is $q(\pi \to \pi') = |A(\pi)|^{-1} |\mathcal{S}(U)|^{-1}$, where $\mathcal{S}(U)$ is the set of admissible splits of U into the two sizes. The reverse move must select the unique pair in π' whose union is U, so $q(\pi' \to \pi) = |A(\pi')|^{-1} |\mathcal{S}(U)|^{-1}$. With the stated acceptance probability, $Q(\pi)q(\pi \to \pi')a(\pi \to \pi') = \min\{Q(\pi)|A(\pi)|^{-1}, Q(\pi')|A(\pi')|^{-1}\}|\mathcal{S}(U)|^{-1}$, which is symmetric in (π, π'), so detailed balance holds. If the proposal returns π itself the move is a self-loop and detailed balance is trivial. The two re-split groups are adjacent because U is connected, and the reverse pair is unique when $\pi' \neq \pi$. Group-locality of the criteria makes $\mathcal{S}(U)$ depend on U alone. Finally, the capital step redraws c from $Q_\theta(c \mid \pi')$ independently of the old capitals, so $Q_\theta(\pi)Q_\theta(c \mid \pi)K(\pi, \pi')Q_\theta(c' \mid \pi')$ is symmetric in $((\pi, c), (\pi', c'))$ whenever $Q_\theta(\pi)K(\pi, \pi')$ is, and the joint kernel is reversible for $Q_\theta(\pi, c)$. Irreducibility is not needed for Theorem 2; it is needed for the simulated design quantities to approximate E_Q, which we check with effective sample sizes. ▫

Proof of Theorem 2. (a) Under the null, $\Delta\widetilde{Y}$ is fixed across maps. By Lemma 1 and the parallel construction, if $P_1 \sim Q_{\theta_0}$ then $\left(P_1, P^{(1)}, \ldots, P^{(S)}\right)$ is exchangeable: P_1 and P_* form a stationary reversible pair, and conditional on P_* all $S+1$ maps are independent m-step transitions from P_* (Besag and Clifford 1989). The rank of $T\left(P_1, \Delta\widetilde{Y}\right)$ among the $S+1$ values is then uniform up to ties, which gives the bound. (b) $\{\hat{p}_{BB} \leq \alpha\} \subseteq \{\theta_0 \notin \Theta_\gamma\} \cup \{\hat{p}\left(\tau^0; \theta_0\right) \leq \alpha - \gamma\}$, whose probability is at most $\gamma + (\alpha - \gamma)$ by (a). (c) With constant nuisance effects, the adjusted outcomes at the true (θ_0, τ_{-k}) are fixed under the null, so (a) applies at that point, and the union bound over a joint $1 - \gamma$ set (or over separate sets with error budgets summing to γ) gives the result. ▫

Lemma 2 (conditional exchangeability). In the construction of Theorem 2(a), let the realized map be drawn from Q^* and let the kernel K be Q-reversible. Then, conditional on the unordered pooled set $\mathcal{M}$,

$\Pr\left(P_1 = P^{(s)} \mid \mathcal{M}\right) \propto Q^*\left(P^{(s)}\right)/Q\left(P^{(s)}\right)$, counting repeated elements separately. In particular, P_1 is uniform on $\mathcal{M}$ when $Q^* = Q$.

Proof. The joint density of $\left(P_1, P_*, P^{(1)}, \ldots, P^{(S)}\right)$ is $Q^*(P_1) K^m(P_1, P_*) \prod_s K^m\left(P_*, P^{(s)}\right)$. Write $Q^*(P_1) = w(P_1) Q(P_1)$ and use reversibility, $Q(P_1) K^m(P_1, P_*) = Q(P_*) K^m(P_*, P_1)$. The density becomes $w(P_1) \cdot Q(P_*) K^m(P_*, P_1) \prod_s K^m\left(P_*, P^{(s)}\right)$. After integrating out P_*, the second factor is a symmetric function of $\left(P_1, P^{(1)}, \ldots, P^{(S)}\right)$. Conditional on the multiset, the probability that the realized map is the element $P^{(s)}$ is therefore proportional to $w\left(P^{(s)}\right)$. ▫

Proof of Theorem 1'. By Lemma 2, $E[g(P_1) \mid \mathcal{M}] = \bar{E}[g]$ for any g, and the projections and the denominator are functions of $\mathcal{M}$ alone. Under A1 to A4, $\Delta Y_i(P) = \Delta Y_i(0) + \sum_l f_{li}(D_{li}(P))$ with $\Delta Y_i(0)$ fixed. Hence $E\left[\sum_i Z_{ki}(P_1) \Delta Y_i \mid \mathcal{M}\right] = \sum_i \{\Delta Y_i(0) \bar{E}[Z_{ki}] + \sum_l \bar{E}[Z_{ki} f_{li}(D_{li})]\}$. The basis contains a constant, so $\bar{E}[Z_{ki}] = 0$. For $l \neq k$, $f_{li}(D_{li})$ is a function of $D_{-k,i}$; split it into its projection on $b\left(D_{-k,i}\right)$, to which Z_{ki} is orthogonal over $\mathcal{M}$ by construction, and the residual r_{li}. That gives (a). Part (b) is Cauchy–Schwarz on each $\bar{E}[Z_{ki} r_{li}]$. For (c), $D_{ki} = Z_{ki} + h_{ki}^{\mathcal{M}}$ and Z_{ki} is orthogonal to $h_{ki}^{\mathcal{M}}$ over $\mathcal{M}$, so $\bar{E}[Z_{ki} D_{ki}] = \bar{E}\left[Z_{ki}^2\right] \geq 0$. With a linear f_{ki}, or a binary D_{ki} (where every f with $f(0) = 0$ is linear), the numerator is $\sum_i \tau_{ki} \bar{E}\left[Z_{ki}^2\right]$. For general absolutely continuous f_{ki}, the representation $f(D) = f(d_0) + \int f'(t) \mathbf{1}\{D \geq t\} dt$ and $\bar{E}[Z_{ki}] = 0$ give the stated weights. ▫

Proof of Proposition 4. By Lemma 2, conditional on $\mathcal{M}$ the realized element is drawn with probabilities $w\left(P^{(s)}\right) / \sum_r w\left(P^{(r)}\right)$. Under the null the values $T\left(P^{(s)}\right)$ are fixed given $\mathcal{M}$, so $\hat{p}_w$ is the upper-tail probability of the realized value under its exact conditional law. The upper-tail probability of a draw from a discrete distribution is super-uniform: $\Pr(\hat{p}_w \leq \alpha \mid \mathcal{M}) \leq \alpha$. Integrating over $\mathcal{M}$ gives the result. ▫

Proof of Theorem 3. Write $T_n = \sum_i Z_{ki} \Delta Y_i = \sum_{c=1}^{G_n} X_c$, where X_c is the total over block c. The X_c are independent under the design by assumption, and $\sup_c E|X_c|^{2+\delta} \leq C$ for some $\delta > 0$, since block sizes and moments are bounded. Let $s_n^2 = Var(T_n) = \sum_c Var(X_c) \geq \kappa G_n$ for some $\kappa > 0$. The Lyapunov ratio satisfies

$$\frac{\sum_c E|X_c - EX_c|^{2+\delta}}{s_n^{2+\delta}} \leq \frac{2^{1+\delta} C G_n}{(\kappa G_n)^{1+\delta/2}} \to 0,$$

so $(T_n - ET_n)/s_n \Rightarrow N(0,1)$. The denominator $\sum_i E_Q[Z_{ki} D_{ki}]$ is a design constant and ET_n is the numerator of β_k by Theorem 1, which gives the stated limit for $\hat{\tau}_k$. For variance estimation, $\widehat{V}_n = \sum_c X_c^2$ has $E\widehat{V}_n = s_n^2 + \sum_c (EX_c)^2 \geq s_n^2$. Its variance is $\sum_c Var\left(X_c^2\right) = O(G_n)$ under 4 moments (or $\widehat{V}_n / E\widehat{V}_n \to 1$ in probability by the weak law for independent arrays under $2 + \delta$ moments), so $\widehat{V}_n / s_n^2 \geq 1 + o_p(1)$, and Wald intervals built on $\widehat{V}_n$ are asymptotically conservative. Weak dependence between

neighbouring blocks with bounded degree is handled by the dependency-graph bounds of Chen and Shao (2004), with $\widehat{V}_n$ summing over blocks and their neighbours. ▫

Proof of Theorem 2(c'). Write $T(P) = \sum_c X_c(P)$ with $X_c(P) = \sum_{i \in c} Z_{ki}(P) \Delta\ddot{Y}_i$. Under H_{0k}, $\Delta Y_i = \Delta Y_i(0) + \sum_{l \neq k} f_{li}(D_{li}(P_1))$, so $T(P_1) = \sum_i Z_{ki}(P_1) \{\Delta Y_i(0) + \sum_{l \neq k} (f_{li}(D_{li}(P_1)) - \hat{\tau}_l D_{li}(P_1))\}$. Its design mean is zero: $E_Q[Z_{ki}] = 0$, $E_Q[Z_{ki} f_{li}(D_{li})] = 0$ for f_{li} in the span of b (Theorem 1'(b) with $R = 0$), and the plug-in term contributes $(\hat{\tau}_l - \beta_l) \sum_i Z_{ki} D_{li}$, which is $o_p\left(G_n^{1/2}\right)$ because $\hat{\tau}_l - \beta_l = O_p\left(G_n^{-1/2}\right)$ by Theorem 3 and $\sum_i Z_{ki} D_{li}$ has mean zero and standard deviation $O\left(G_n^{1/2}\right)$. By Theorem 3, $T(P_1)/s_n \Rightarrow N(0,1)$ and $\widehat{V}(P_1)/s_n^2 \geq 1 + o_p(1)$, so $t(P_1) \Rightarrow N(0, \rho^2)$ with $\rho \leq 1$ along subsequences. For the reference distribution, hold $a = \Delta\ddot{Y}$ fixed: under draws from Q, $\sum_i Z_{ki}(P) a_i$ is a sum of independent block totals with mean zero, since $E_Q[Z_{ki}] = 0$ for every unit. Its studentised version is therefore asymptotically $N(0,1)$, and the empirical quantiles over the companions converge to the normal quantiles (Chung and Romano 2013; Wu and Ding 2021). Rejecting when $t(P_1)$ falls outside those quantiles then has asymptotic probability at most α because $\rho \leq 1$. ▫